

DRUGS USED IN GOUT

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OVERVIEW

- Gout is a familial metabolic disease characterised by recurrent episodes of acute arthritis due to deposits of monosodium urate crystals in joints and cartilage
- Uric acid calculi can also occur in the kidneys
- Usually, it is usually associated with high serum levels of uric acid which is a poorly soluble and major end product of purine metabolism
- In most mammals, the enzyme uricase is responsible for converting uric acid to the more soluble allantoin, however this enzyme is absent in humans

PATHOPHYSIOLOGY OF GOUT

- Urate crystals are initially phagocytosed by synoviocytes which then release prostaglandins, lysosomal enzymes and interleukin - 1
- These attract and activate polymorphonuclear leukocytes and mononuclear phagocytes
- Attracted by chemotactic mediators, these phagocytes migrate into the joint space and amplify the ongoing inflammatory process
- Later in the attack, there is increased number of phagocytes which ingest the urate crystals and release more inflammatory mediators
- From this it can be deduced that the most effective medications for management of acute urate crystal induced inflammation should suppress the different phases of leukocyte activation

HYPERURICEMIA NOT ALWAYS EQUAL TO GOUT

- Patients with hyperuricemia associated to gout and urate crystal formation must be clearly distinguished from those with only hyperuricemia
- This is because treatment of hyperuricemic patients without symptoms of gout has no long term approved efficacy
- Therefore treatment of such is not recommended unless the association between hyperuricemia and gout is established

AIMS OF GOUT TREATMENT

- Relieving the acute gout attack
- Prevent reformation urate crystals
- Subsequent prevention of recurrence of gout episodes

GENERAL CLASSIFICATION OF DRUGS USED IN GOUT

- Drugs used for treatment of gout fall under the following categories;
 1. Inhibitors of uric acid synthesis e.g allopurinol
 2. Uricosuric (excretion enhancers) drugs e.g. probencid, sulfinpyrazone
 3. Inhibitors of inflammatory mediators e.g. colchicine, NSAIDs, Corticosteroids

1. COLCHICINE

- It is neither analgesic nor anti-inflammatory but suppresses gout inflammation
- It does not inhibit the synthesis neither does it promote excretion of uric acid nor affect blood uric acid levels

Pharmacokinetics of Colchicine

- Absorbed readily after its administration and reaches peak plasma levels within 2 hours
- Metabolites are excreted through the GIT and urine

Pharmacodynamics of Colchicine

- Relieves pain and inflammation of gout in 12 to 24 hours without altering the metabolism or excretion of urates

Mechanism of Action of Colchicine

- Prevents the release of chemotactic factors and inflammatory cytokines from neutrophils and thus reducing attraction of more neutrophils in the affected joint area
- Produces anti-inflammatory effects by binding to the intracellular protein tubulin thereby preventing polymerisation to microtubules and thus leading to inhibition of leukocyte migration and phagocytosis
- Colchicine adverse effects are produced by its inhibition of tubulin polymerisation and cell

INDICATION OF COLCHICINE

- Acute gout arthritis

Adverse effects

- Diarrhoea, nausea, vomiting and abdominal pain, rarely hair loss, bone depression and peripheral neuritis

Acute Colchicine toxicity

- Burning throat pain, bloody diarrhea, hematuria and oliguria, nervous system depression which is managed through supportive treatment

2. NSAIDS

- Indomethacin is the NSAID of choice as unlike Acetyl salicylic acid, it does not compete with uric acid for the secretory or excretory pathway
- Apart from inhibition of prostaglandin synthesis, Indomethacin and other NSAIDs apart from aspirin also inhibits the urate crystal phagocytosis
- Indomethacin is commonly used as initial treatment of gout

3. CORTICOSTEROIDS

- Often suggested for elderly patients with chronic tophaceous gout
- Intra-articular injections mostly given in those not tolerating NSAIDs or colchicine

4. URICOSURIC AGENTS

- Example; probenecid, sulfinpyrazone
- Uricosuric drugs are also referred to as urate diuretics
- have a similar structure to urate structure and hence compete with uric acid for transport sites
- High doses increases uric acid excretion by inhibiting proximal tubular reabsorption

GUIDANCE ON INITIATING OF URICOSURIC AGENTS

- Initial phase of uricosuric agents is the most dangerous period in that it initially causes increase in uric acid
- These drugs have first to build to sufficient concentration before they can effectively start to inhibit uric acid reabsorption and secretion
- This causes increased chances of acute gout attack and renal stone formation
- To prevent this, it is recommended that small amounts of Colchicine be given before adding uricosuric drugs on the therapeutic regimen
- It is worthy noting that Probenecid was initially developed to prolong penicillin blood levels

INDICATIONS OF URICOSURIC DRUGS

- Initiated if several acute attacks of gout arthritis have occurred
- When there is evidence of tophi
- When plasma levels of uric acid are so high in patients with gout

NB:

- Therapy of uricosuric agents should not be started a number of weeks (2 to 3 weeks) after an acute attack

SIDE EFFECTS

- GIT irritation
- probencid causes allergic dermatitis
- Nephrotoxic syndrome
- Aplastic anemia though rare

CONTRAINDICATION AND CAUTIONS AGAINST URICOSURIC AGENTS

- The drugs are contraindicated in patients with history of renal calculi
- Essential to maintain a large urine volume to minimise the possibility of stone formation

5. URIC ACID SYTHESIS INHIBITORS

- Allopurinol is the drug of choice in treatment of chronic tophaceous gout
- It is especially useful in patients whose treatment is complicated by renal insufficiency
- Inhibition of uric acid sythesis by inhibiting xanthine oxidase makes allopurinol as an alternative to increasing uric acid excretion in the treatment

MECHANISM OF ACTION OF ALLOPURINOL

- Nucleic acids are converted to xanthines or hypoxantines and then oxidised to uric acid
- This process is facilitated by the enzyme xanthine oxidase
- Inhibition of this enzyme leads to reduced urinary and blood concentration of uric acid
- There is further increase in excretion of the more water soluble uric acid precursors, xanthines and hypoxanthine

PHARMACOKINETICS OF ALLOPURINOL

- Approximately 80% of drug is absorbed after oral administration
- Like uric acid, allopurinol is also metabolised by xanthine oxidase
- Alloxantine, the resulting compound has the capacity to inhibit xanthine oxidase and has a long enough duration of action
- This explains why allopurinol need to be given only once per day

INDICATIONS OF ALLOPURINOL

- In chronic tophaceous gout in which reabsorption of tophi is more rapid than uricosuric agents
- In gout patients whose 24hour urinary uric acid remains high despite being on purine free diet
- When probenecid or sulfinpyrazone cannot be used because of adverse effects or allergic reactions or when proved to effective
- Recurrent renal stones
- Patients with renal impairment
- When serum urates are grossly elevated

ADVERSE EFFECTS

- GIT intolerance such as nausea, vomiting and diarrhoea
- Peripheral neuritis and necrotising vasculitis, depression of bone marrow elements and rarely aplastic anemia may occur
- Allergic reactions
- Rare cases of cataracts due to the binding of allopurinol to the lenses

INTERACTIONS AND CAUTIONS AGAINST ALLOPURINOL

- Inhibits metabolism of probencid and oral anticoagulants
- Safety in children and during pregnancy has not been established

END